

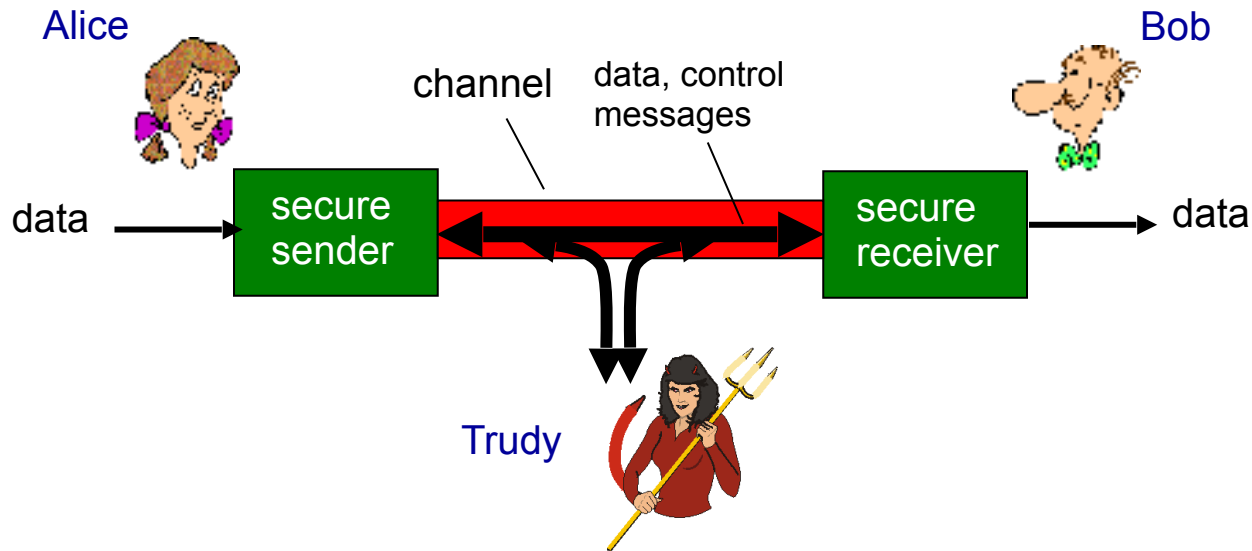
CS151: Encryption and blockchain



Speaker: Xiaoxue (Eira) Zhang

Friends and enemies: Alice, Bob, Trudy

- well-known in network security world
- Bob, Alice (lovers!) want to communicate "securely"
- Trudy (intruder) may intercept, delete, add messages



Who might Bob, Alice be?

- ... well, real-life Bobs and Alices!
- Web browser/server for electronic transactions (e.g., on-line purchases)
- on-line banking client/server
- DNS servers
- other examples?

There are bad guys (and girls) out there!

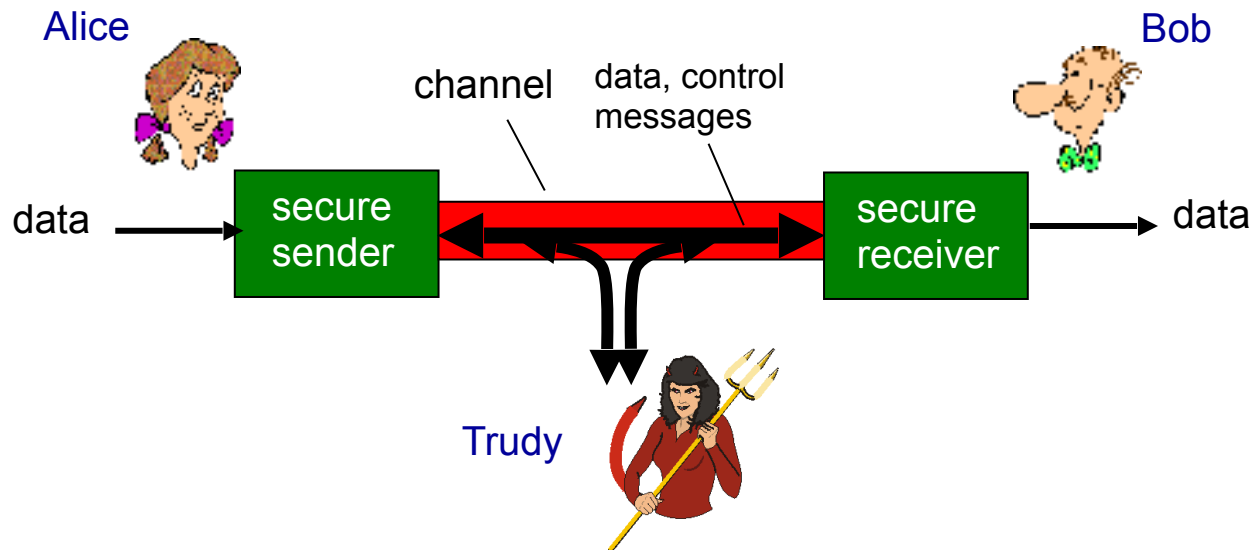
Q: What can a “bad guy” do?

A: A lot!

- **eavesdrop**: intercept messages
- actively **insert** messages into connection
- **impersonation**: can fake (spoof) source address in packet (or any field in packet)
- **hijacking**: “take over” ongoing connection by removing sender or receiver, inserting himself in place
- **denial of service**: prevent service from being used by others (e.g., by overloading resources)

Secure communication

- Confidentiality: Keep data and communication secret
 - Encryption / decryption
- Authenticity: Parties cannot be impersonated
 - "Did Alice really send this message?"
- Integrity: Messages cannot be modified
 - "Was this the original message that was sent?"



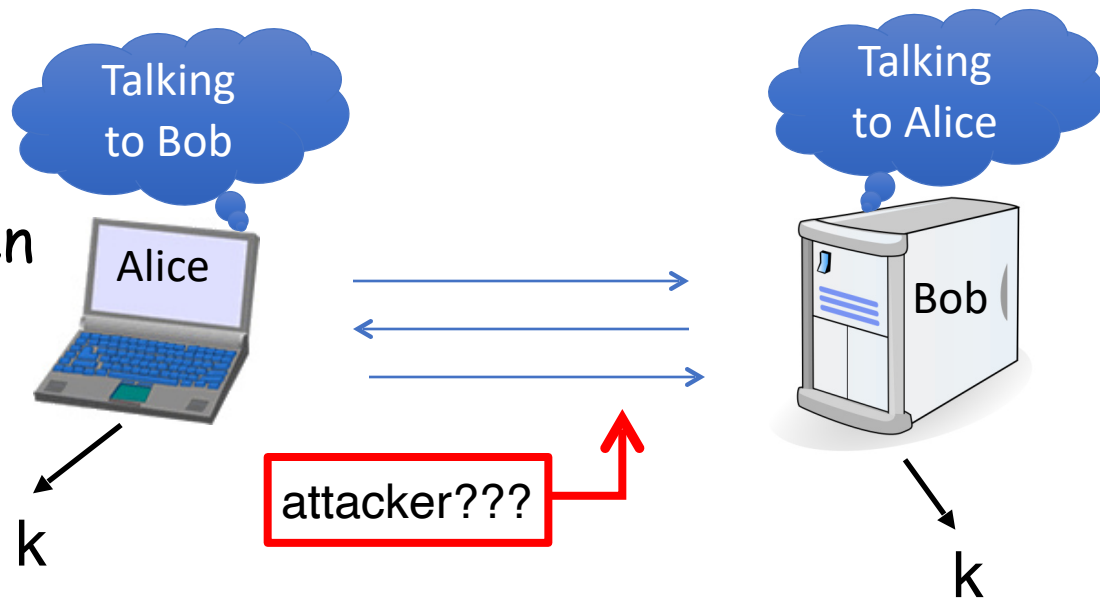


Introduction

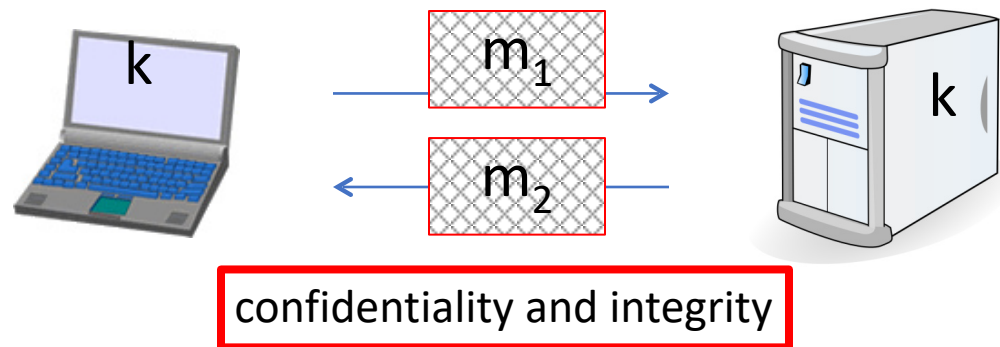
What is cryptography?

Core of Cryptography

Secret key establishment



Secure communication:



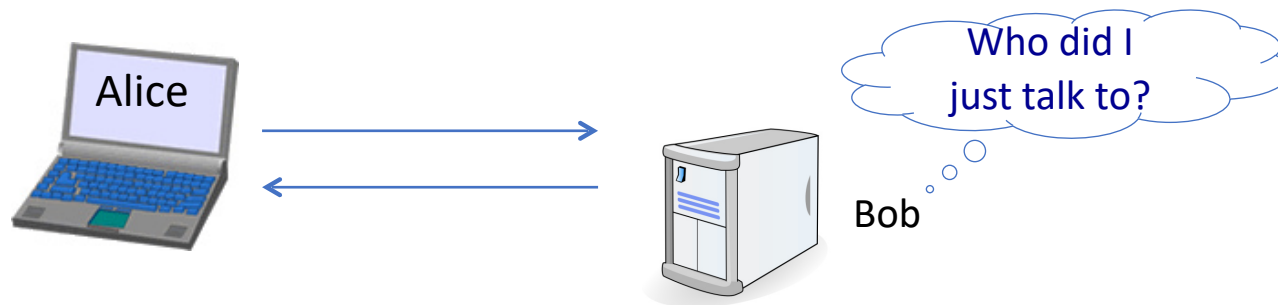
But crypto can do much more

- Digital signatures
 - Sender (Bob) digitally signs document, establishing he is the document owner/creator
 - Make the digital signature via function of the content being signed
 - Verifiable, non-forgeable
 - recipient (Alice) can prove to someone that Bob, and no one else (including Alice), has signed document



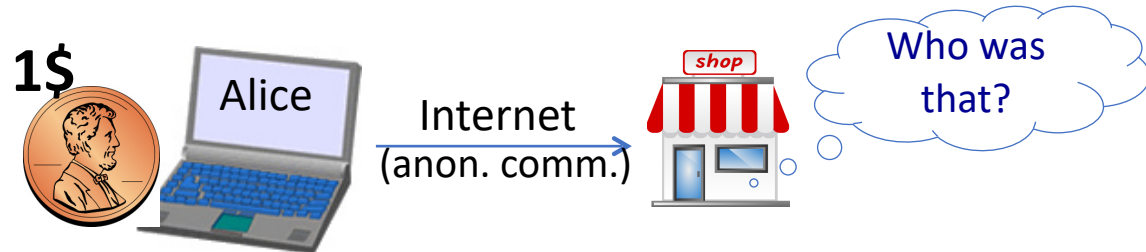
But crypto can do much more

- Anonymous communication
 - Mix network: Alice sends messages to Bob through a sequence of proxies
 - Messages are encrypted and decrypted appropriately
 - Bob doesn't know who is the sender



But crypto can do much more

- **Anonymous digital cash**
 - Can I spend a “digital coin” without anyone knowing who I am?
 - How to prevent double spending?





Introduction

History

The dancing men

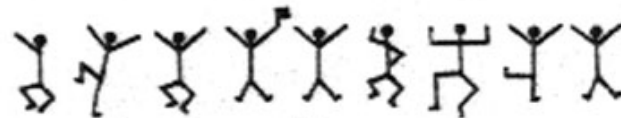
- A story about Sherlock Holmes, by Sir Arthur Conan Doyle.



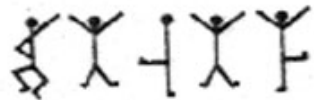
The dancing men



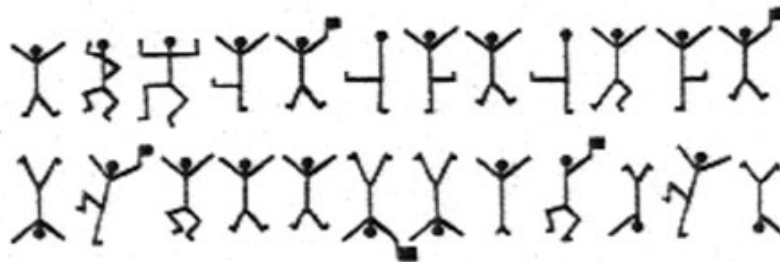
criminal's message (1)



criminal's message (2)



Elsie's reply



criminal's message (3)

Few Historic Examples (all badly broken)

Substitution cipher: substituting one thing for another

$$c := E(k, "bcza") = "wnac"$$

$$D(k, c) = "bcza"$$

$k :=$

a	$\rightarrow$	c
b	$\rightarrow$	w
c	$\rightarrow$	n
$\vdots$		
z	$\rightarrow$	a

Key: substitution table

How to break a substitution cipher?

(1) Use frequency of English letters

“e”: 12.7%, “t”:9.1%, “a”: 8.1%

(2) Use frequency of pairs of letters (digrams)

common pairs: “he”, “an”, “in”, “th”

ciphertext-only attack

An Example

UKBYBIPOUZBCUFEEBORUKBYBHOBBERFESPVKBWFOFERNBCVBZPRUBOFE
RVNBCVBPCYYFVUFOFEIKNWFRFIKJNUPWRFIPOUNVNIPUBRNCUKBEFWWFD
NCHXCYBOHOPYXPUBNCUBOYNRVNIWNCPOJIOFHOPZRVFZIXUBORJRUBZR
BCHNCBBONCHRJZSFWNVJRUBZRPCYZPUKBZPUNVPWPCYVFZIXUPUNFCP
WRVNBCVBRPYYNUNFCPWWJUKBYBIPOUZBCUIPOUNVNIPUBRNCCHOPYXPUB
NCUBOYNRVNIWNCPOJIOFHOPZRNCRVNBCUNENVVFZIXUNCHPCYVFZIXUPU
NFCPWZPUKBZPUNVR

B	36	→ E
N	34	
U	33	→ T
P	32	→ A
C	26	

NC	11	→ IN
PU	10	→ AT
UB	10	
UN	9	

digrams

UKB	6	→ THE
RVN	6	
FZI	4	

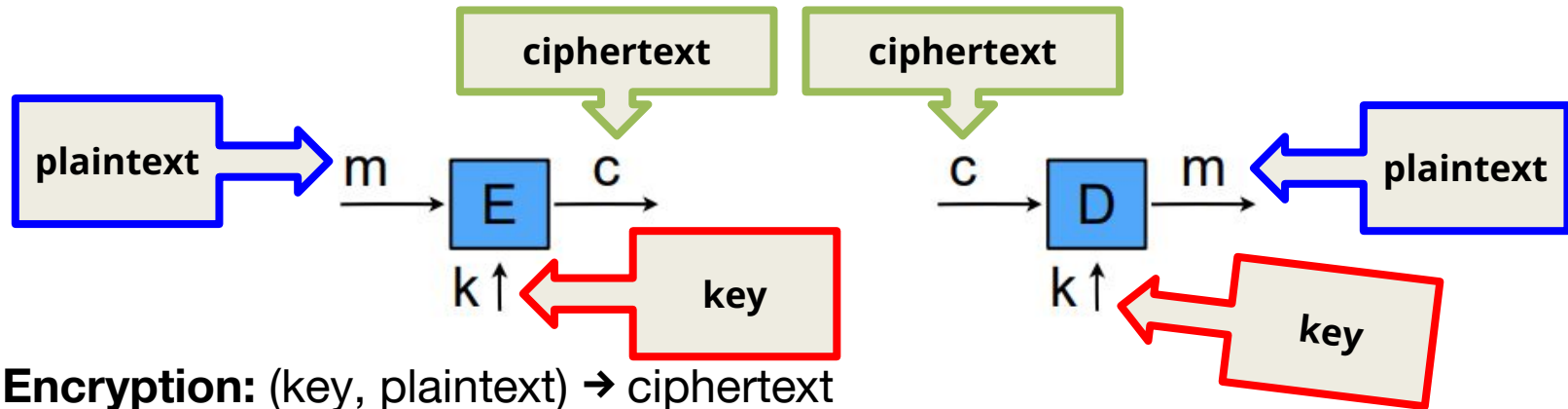
trigrams

Symmetric-key encryption



- **Encryption:** (key, plaintext) $\rightarrow$ ciphertext
 - $E_k(m) = c$
- **Decryption:** (key, ciphertext) $\rightarrow$ plaintext
 - $D_k(c) = m$
- **Functional property:** Where $D_k(E_k(m)) = m$

Symmetric-key encryption



- **Encryption:** (key, plaintext) $\rightarrow$ ciphertext

➤ $E_k(m) = c$

- **Decryption:** (key, ciphertext) $\rightarrow$ plaintext

➤ $D_k(c) = m$

- **Functional property:** Where $D_k(E_k(m)) = m$

The One Time Pad (Vernam 1917)

$$c := E(k, m) = k \oplus m$$

$$m := D(k, c) = k \oplus c$$

msg:	0	1	1	0	1	1	1	⊕
key:	1	0	1	1	0	1	0	
CT:	1	1	0	1	1	0	1	

$$D(k, E(k, m)) = D(k, k \oplus m) = k \bigoplus (k \oplus m) = m$$

The One Time Pad (Vernam 1917)

You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?

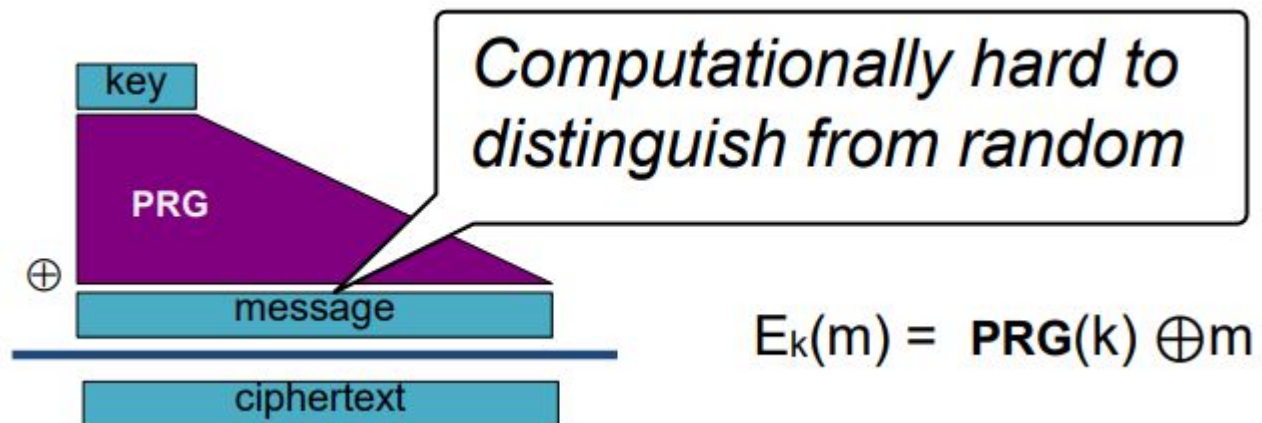
Yes, the key is $k = m \oplus c$.

The One Time Pad (Vernam 1917)

- Shannon (1949)
 - Information-theoretic security: without key, ciphertext reveals no “information” about plaintext
- Problems with OTP
 - Can only use key once
 - Key is as long as the message
 - No integrity protection

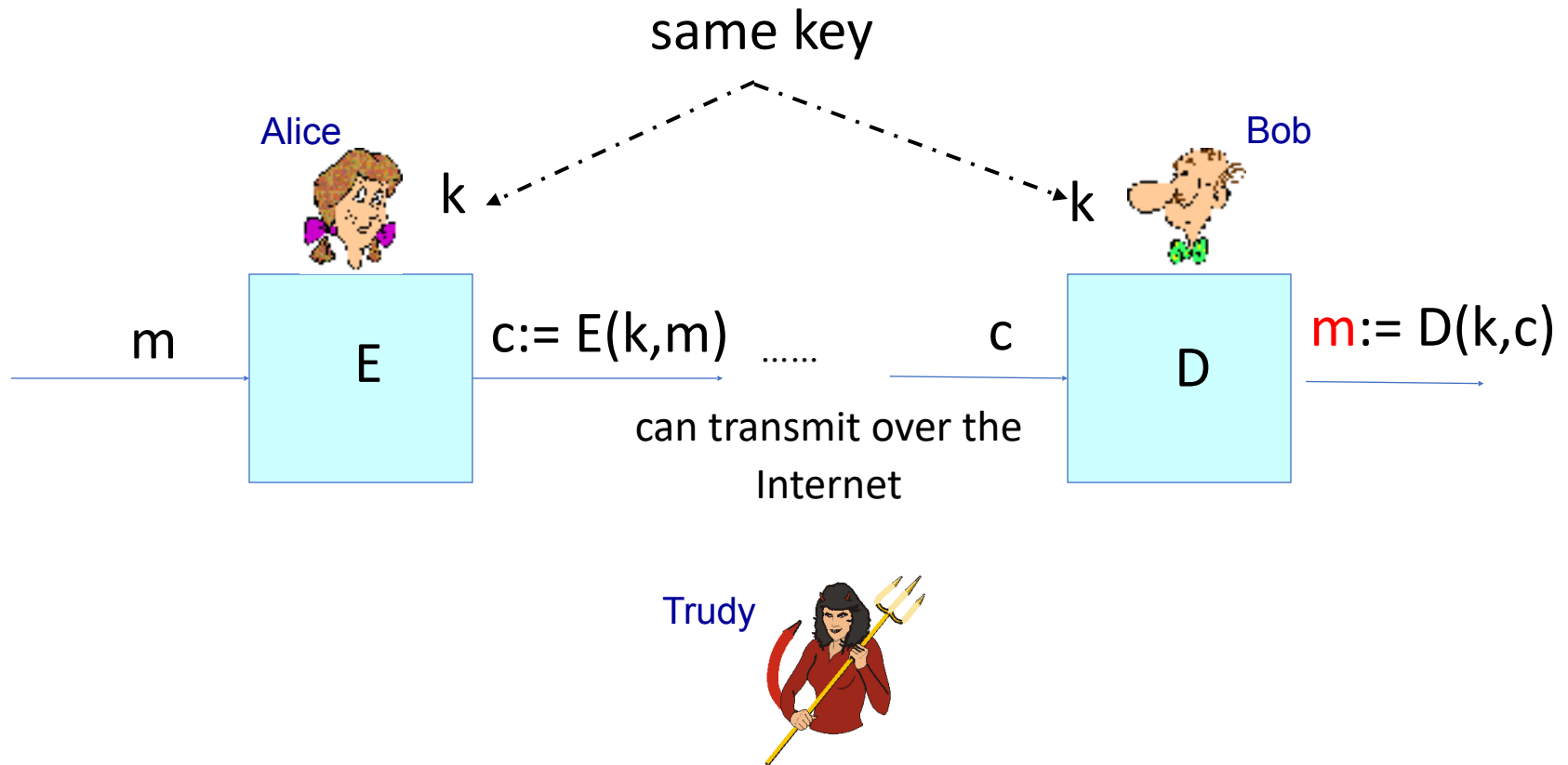
Stream Ciphers: making OTP practical

- Problem: OTP key is as long as message
- Solution: Pseudo random generator
- Stream ciphers uses bit-by-bit to generate ciphertext.

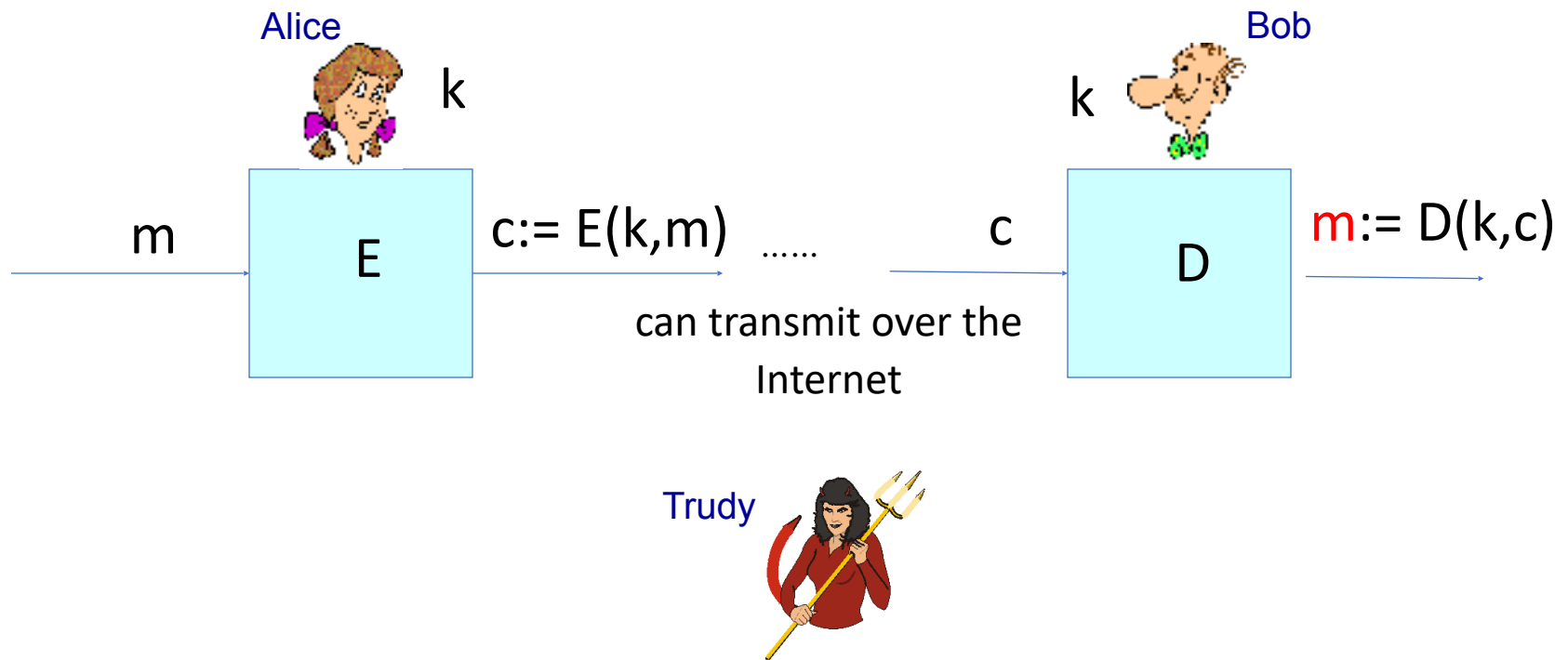


Examples: ChaCha, Salsa, etc.

Symmetric Ciphers



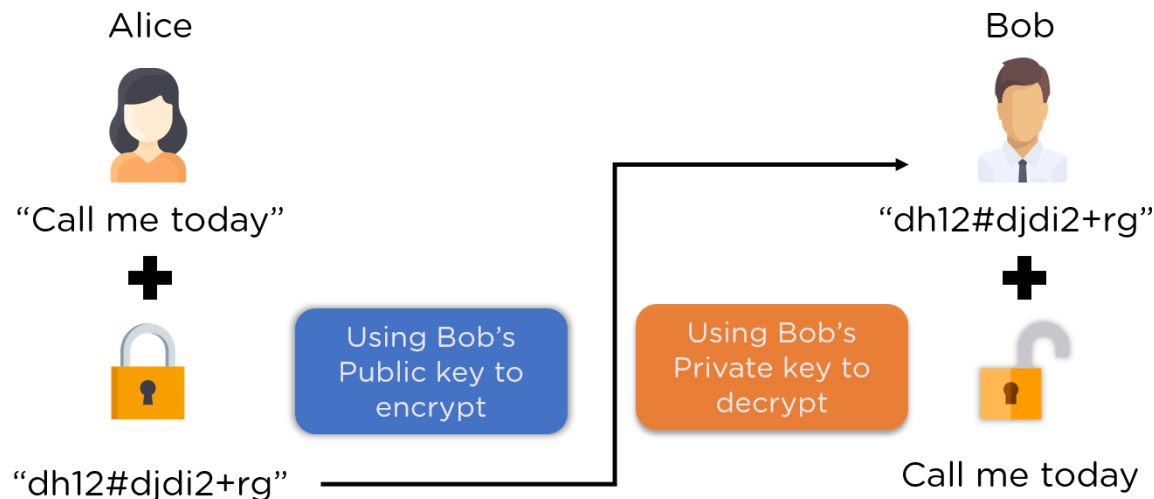
Main challenge: How to share K safely to begin with?



Asymmetric encryption

Use two different keys:

- Public Key - shared with everyone
- Private Key - kept secret by the owner



Asymmetric encryption

Authentication

simple digital signature for message m :

- Alice signs m by encrypting with her private key K_A^- , creating "signed" message, $K_A^-(m)$

Alice's message, m

Dear Bob
Oh, how I have missed
you. I think of you all the
time! ... (blah blah blah)
Alice



K_A^- Alice's private
key

Public key
encryption
algorithm

$m, K_A^-(m)$

Alice's message, m ,
signed (encrypted)
with her private key

From Encryption to Blockchain

Cryptography gave us the tools to:

- Keep data secret (encryption)
- Verify identity (digital signatures)
- Detect tampering (hashing)

Blockchains combine these tools to solve a new problem:

How do we keep a shared database secure when no one is fully trusted?

Six reasons to care about Blockchains

Reason 1: You like excitement!



Source: CNBC

45 BTC = Lambo

Reason 1: You like excitement!



Reason 2: You like depression



Reason 3: You like rollercoasters

Market Summary > Bitcoin

102,701.40 USD

+5,077.48 (5.20%) ↑ past month

Jan 21, 3:19 AM UTC · [Disclaimer](#)

\$60,000.00

\$40,000.00

\$20,000.00

\$0.00

Down 66%

Up 100%



2014

2015

2016

2017

2018

2019

2020

2021

2022

2023

2024

Reason 4: You have a sense of the absurd



 **DENTAC**
The First Blockc
Solution for the G
Dental Indust



**Trump promotes new meme coin
before taking office on pro-crypto
agenda**



Reason 5: Everyone else cares

Forbes

Insurance Interrupted: How Blockchain Innovation is Transforming the Insurance Industry



Andrea Tinianow Contributor @
Crypto & Blockchain
I am the Blockchain Czarina. I bring y

3,686 views | Dec 19, 2018, 07:10am

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David Petersson Contributor @
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Joshua Fraser CommunityVoice
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Real Estate

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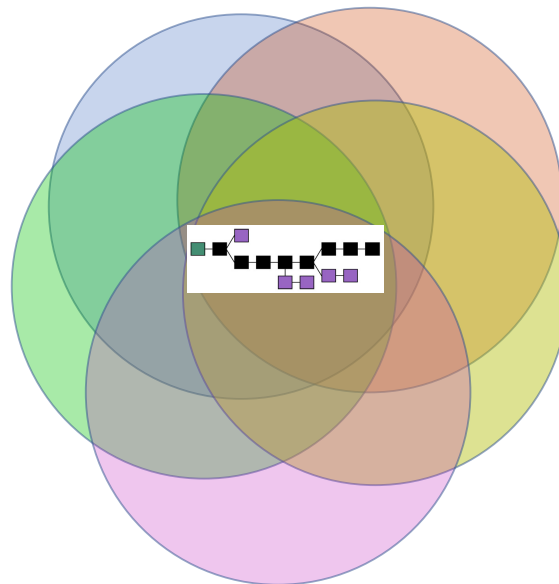
Joshua Fraser

CEO at Data Nerds and Estatic. Focused on bringing transparency into residential/commercial real estate through data and analytics.

Reason 6: They are a fascinating technology spanning many areas of computer science-and beyond

For example:

- Distributed systems
- Programming languages
- Formal verification
- Cryptography / security
- Game theory
- Technology law
- Business



Background

- Centralized trading system



Background

- Centralized DNS

A trusted third party



Background

- Centralized PKI

A trusted third party



Background

❑ Problem

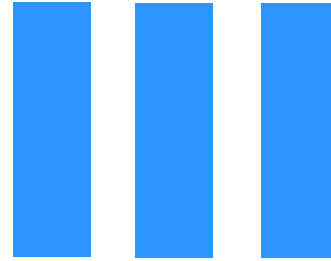
- Single point of failure

❑ What if the system is decentralized and there is no trusted third party?

- We need a distributed ledger
(blockchain)

What is a blockchain?

Blockchain



- ❑ It looks like a file
- ❑ Append-only global log
- ❑ Every node on the network has a consistent copy

What is a blockchain?

- A distributed system of computers...

 **Anyone** can join the protocol (Permissionless)

- running on a peer-to-peer network...
- with the ability to run arbitrary user programs (smart contracts)...
- that can program money/assets.

 **Not** owned or controlled by any company or country. (Decentralized)

 Very strong security, reliability, and neutrality.

   A trustworthy “world computer” for everyone.

Some blockchain use cases

Platform for digital assets (**cryptocurrency**) with growing adoption

- 40% of American adults now own “crypto” (report by security.org)
- Bitcoin and Ethereum ETF (Exchange-Traded Funds)

Statement on the Approval of Spot Bitcoin Exchange-Traded Products

[Chair Gary Gensler](#)

Jan. 10, 2024



**U.S. Securities and
Exchange Commission**

Will Schmitt in New York JULY 22 2024

FINANCIAL TIMES

**SEC approves ether ETFs as crypto moves
closer to mainstream**

Latest authorisations follow January launches of first US spot bitcoin ETFs

Some blockchain use cases

Global aid & remittance payments:

- \$225M in cryptocurrency donations sent to Ukraine
- OFAC-authorized humanitarian aid to 60,000 healthcare workers in Venezuela, using USDC and VPNs evaded interception by the local regime

Ukraine Bought Weapons, Drones With Crypto Donations

The wartorn nation also revealed it bought non-lethal equipment in a new report detailing its expenditures from crypto donations.

By Amitoj Singh ⌚ Aug 17, 2022 at 12:17 p.m. EDT Upd:

Circle Partners with Bolivarian Republic of Venezuela and Airtm to Deliver Aid to Venezuelans Using USDC

<https://www.circle.com/blog/circle-partners-with-bolivarian-republic-of-venezuela-and-airtm-to-deliver-aid-to-venezuelans-using-usdc>

Some blockchain use cases

DeFi (Decentralized Finance): Financial systems run on “blockchain computer”

- Users *don't* need to trust companies for their funds
- Auditing can be facilitated by *formal verification* of smart contract code
- Open interoperability & innovation



"with some simple tweaks to computer code," [SBF] helped Alameda Research **misappropriate as much as \$65 billion** from FTX customers.

Smart-contract-powered Decentralized Exchanges won't have the same vulnerability!

Some blockchain use cases

- Platform for data recording/tracking
 - Identity/Naming systems
 - Access-control systems
 - ...



A more human passport for the Internet. More powerful. More integrations.



California DMV puts 42 million on blockchain to fight fraud

By Akash Sriram

July 30, 2024 6:59 PM EDT · Updated a month ago

Handshake is a decentralized, permissionless naming protocol where every peer is validating and in charge of managing the root DNS naming zone with the goal of creating an alternative to existing Certificate Authorities and naming systems. Names on the

Blockchain's Origin



Bitcoin Genesis Block

Raw Hex Version

```
00000000 01 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000010 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000020 00 00 00 00 3B A3 ED FD 7A 7B 12 B2 7A C7 2C 3E .....;f1yz{.²zÇ,>
00000030 67 76 8F 61 7F C8 1B C3 88 8A 51 32 3A 9F B8 AA gv.a.B.Ã~$Q2:Ÿ,*
00000040 4B 1B 5E 4A 29 AB 5F 49 FF FF 00 1D 1D AC 2B 7C K.^J)«_iÿŸ...¬+|
00000050 01 01 00 00 00 01 00 00 00 00 00 00 00 00 00 .....
00000060 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 .....
00000070 00 00 00 00 00 00 FF FF FF FF 4D 04 FF FF 00 1D .....ÿÿÿÿM.ÿÿ..
00000080 01 04 45 54 68 65 20 54 69 6D 65 73 20 30 33 2F ..EThe Times 03/
00000090 4A 61 6E 2F 32 30 30 39 20 43 68 61 6E 63 65 6C Jan/2009 Chancel
000000A0 6C 6F 72 20 6F 6E 20 62 72 69 6E 6B 20 6F 66 20 lor on brink of
000000B0 73 65 63 6F 6E 64 20 62 61 69 6C 6F 75 74 20 66 second bailout f
000000C0 6F 72 20 62 61 6E 6B 73 FF FF FF FF 01 00 F2 05 or banksÿÿÿÿ..ð.
000000D0 2A 01 00 00 00 43 41 04 67 8A FD B0 FE 55 48 27 *....CA.g$Ÿ"bUH'
000000E0 19 67 F1 A6 71 30 B7 10 5C D6 A8 28 E0 39 09 A6 .gã|g0·.\ð"(à9.|
000000F0 79 62 E0 EA 1F 61 DE B6 49 F6 BC 3F 4C EF 38 C4 ybâe.a>9Iðk?L18Ã
00000100 F3 55 04 E5 1E C1 12 DE 5C 38 4D F7 BA 0B 8D 57 6U.Ã.ð\8M+9..W
00000110 8A 4C 70 2B 6B F1 1D 5F AC 00 00 00 00 ŠLp+kã._¬....
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Blockchain

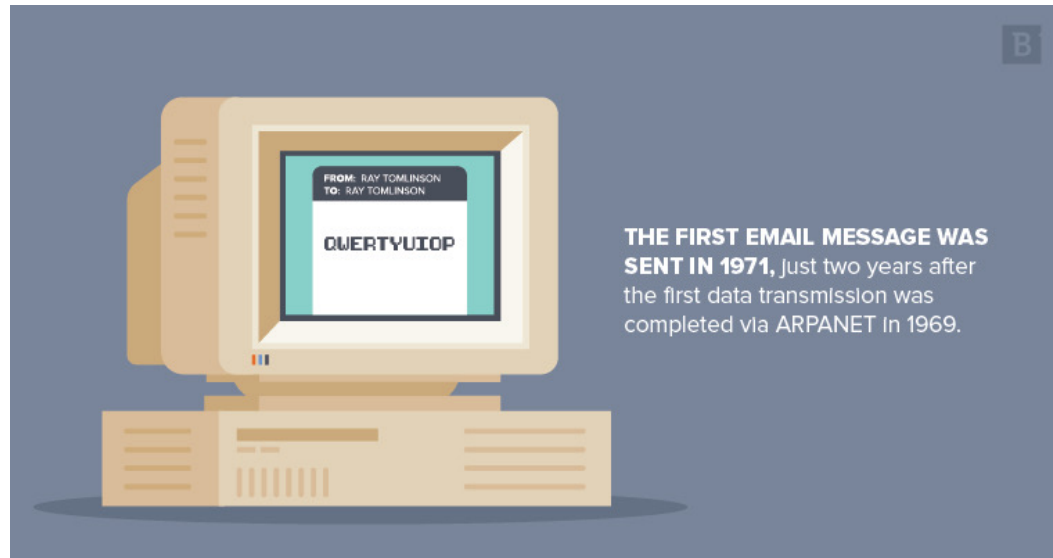


Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto
satoshin@gmx.com
www.bitcoin.org

Abstract. A purely peer-to-peer version of electronic cash would allow online payments to be sent directly from one party to another without going through a financial institution. Digital signatures provide part of the solution, but the main benefits are lost if a trusted third party is still required to prevent double-spending. We propose a solution to the double-spending problem using a peer-to-peer network. The network timestamps transactions by hashing them into an ongoing chain of hash-based proof-of-work, forming a record that cannot be changed without redoing the proof-of-work. The longest chain not only serves as proof of the sequence of events witnessed, but proof that it came from the largest pool of CPU power. As long as a majority of CPU power is controlled by nodes that are not cooperating to attack the network, they'll generate the longest chain and outpace attackers. The network itself requires minimal structure. Messages are broadcast on a best effort basis, and nodes can leave and rejoin the network at will, accepting the longest proof-of-work chain as proof of what happened while they were gone.

Time travel to 1971



Fun fact: Email systems are early examples of decentralized systems.

- You'd think "hmm, it would be great if I can send \$\$\$ over e-mail"
- How would you do it?

First attempt

- Take a picture of your banknote & send money.jpeg as email attachment?

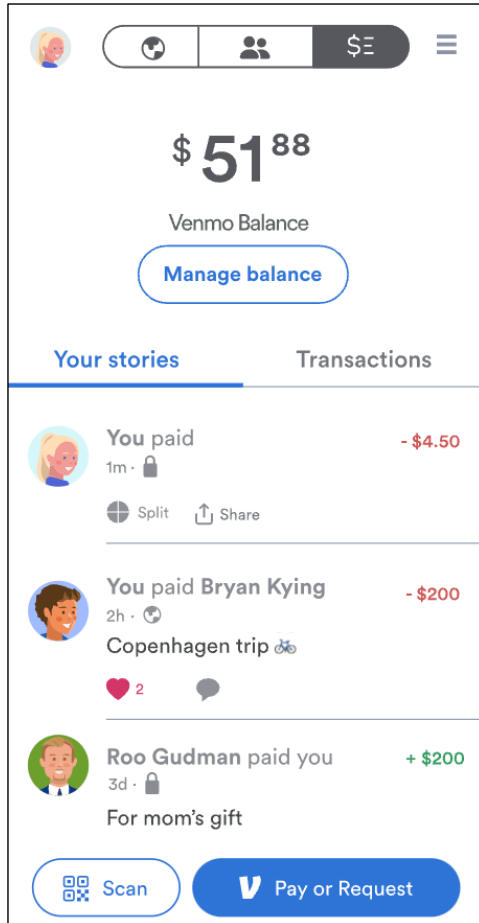


Seems like we need a central party

- Assume a central party accessible by all
- This party is a Trusted Third Party (TTP)
- Rough idea (v1)
 - User sets up account with TTP and deposits \$\$\$
 - Depositing can be done by mailing a check
 - For Alice to pay Bob, she sends an email to TTP, CCing Bob. TTP moves numbers on its balance book.
 - Withdraw: TTP sends a check.
- Double spending solved!
 - Because TTP is involved in all transactions.



We successfully enabled sending money over emails!



- We are all familiar with the modern version of this idea: PayPal, Venmo, WeChat Pay, Zelle, etc
- With much improved user experiences
 - E.g., Email interaction replaced w/ HTTP
- But the core idea is the same...

Pros / Cons of using TTP (v1)

Pros:

- Dirt simple!
- Fast transactions
- TTP can fix errors / reverse transactions

Cons:

- No privacy from TTP
- TTP can cheat users



Bitcoin: A Peer-to-Peer Electronic Cash System

Satoshi Nakamoto

[1] Nakamoto, Satoshi. "Bitcoin: A peer-to-peer electronic cash system." (2008): 28.

Motivation

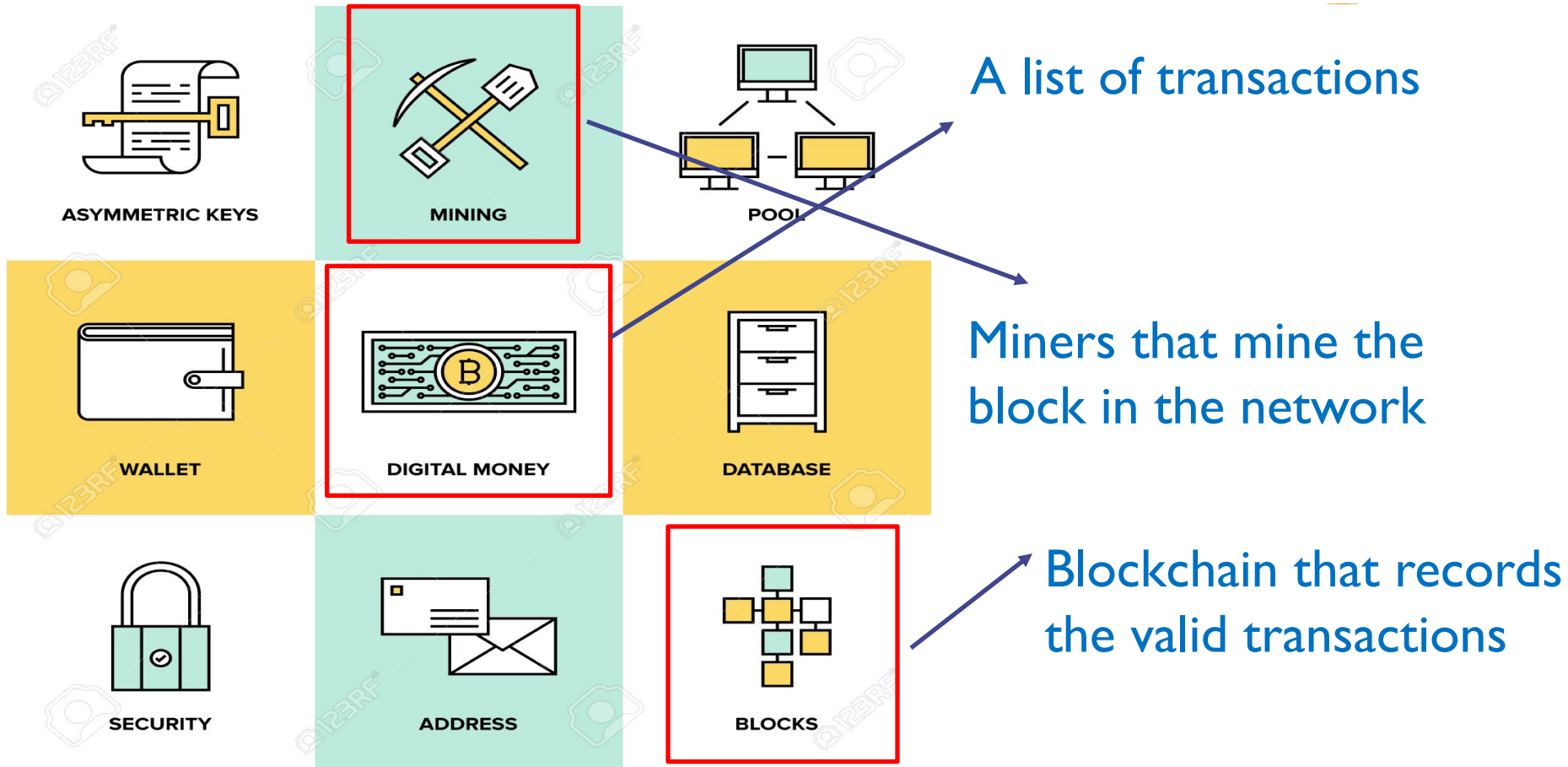


- ◆ Design a decentralized system to conduct direct transaction between any two parties
- ◆ Solve the **double-spending problem**
- ◆ Rely on proof instead of trust
 - **Everyone can't be trusted in the network**



BLOCKCHAIN

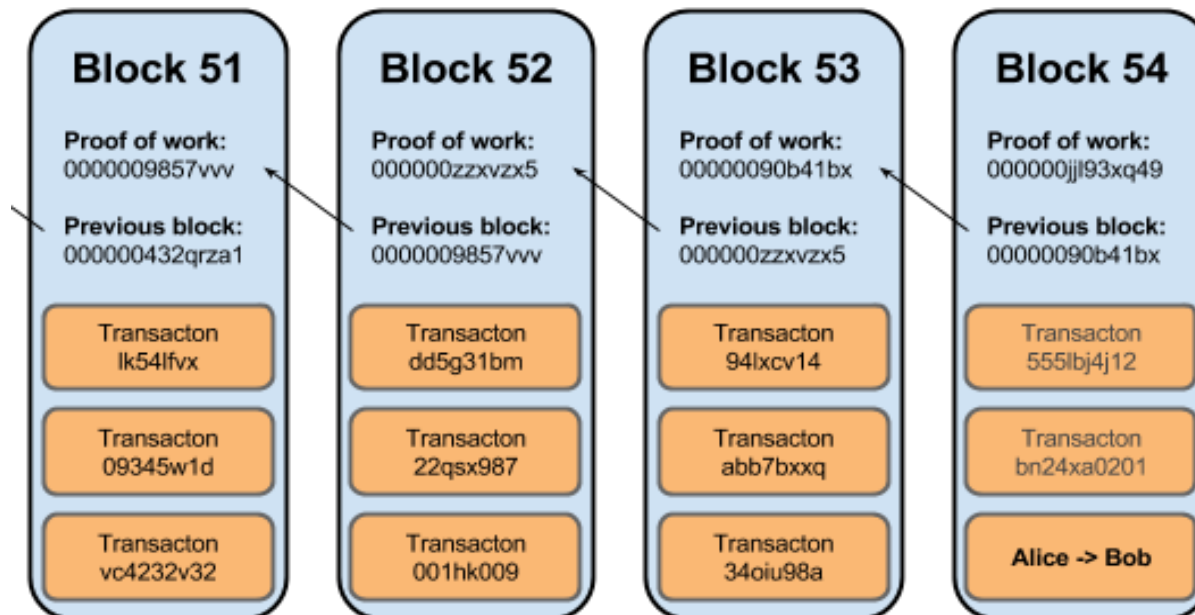
Bitcoin components



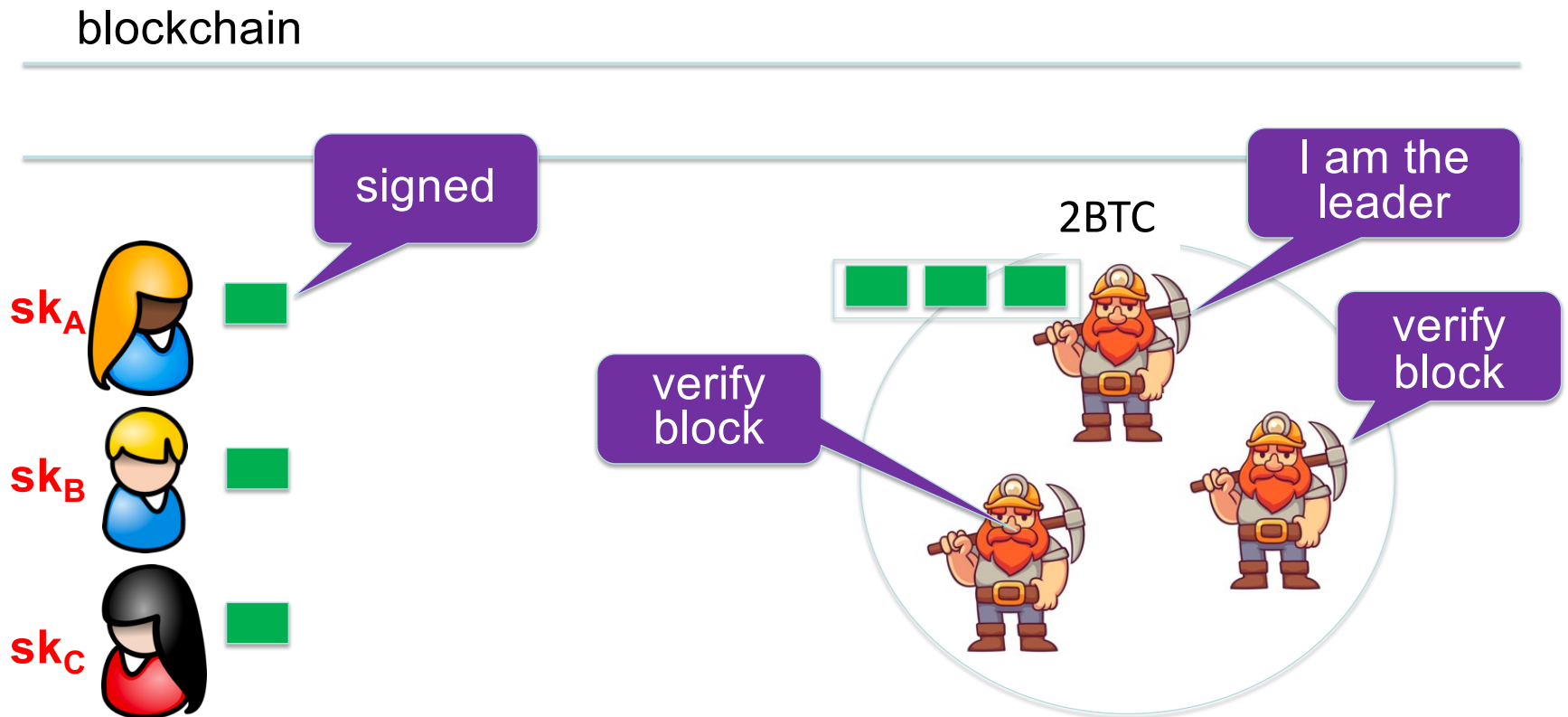
Blockchain



- ◆ Transactions are organized as blocks
- ◆ Blocks are linked
- ◆ Verify every transaction when adding a transaction into a block.



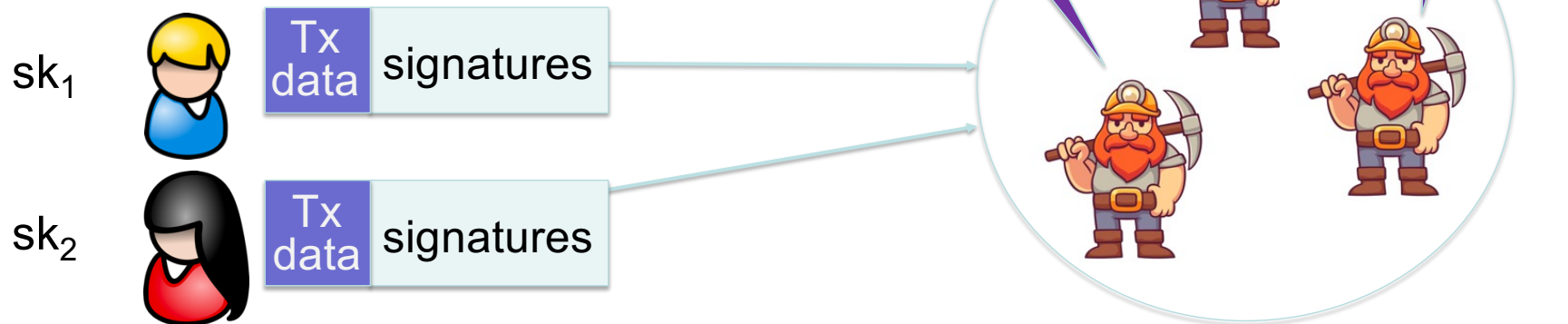
Add Block to the Chain



Verify transactions

Signatures are used everywhere:

- ensure Tx authorization,
- governance votes,
- consensus protocol votes.



Bitcoin

- Pseudonymous: User identities correspond to (SK, PK) key pairs
- Users digitally sign transactions to authorize movement of money
- Transactions recorded in blockchain

Bitcoin

- Except that the blockchain is *fully decentralized*
- Instead of *one* trusted entity, we rely on *whole community*
- Community maintains fully public blockchain / ledger, so that...
- No bank or jurisdiction can suppress transactions

Any Questions?

